Lecture 5

Characters

Lecture 4 decomposed a representation by hunting for invariant subspaces. This lecture replaces the hunt by arithmetic. The character of a representation is the trace of its matrices, one number per group element. Averaging over the group, the argument of Lecture 4, shows that the characters of the irreducible representations are orthonormal. From this follows everything else: characters count multiplicities, test irreducibility, determine a representation, and fix the dimensions of the irreducible representations. We build the character table of $C_{3v} \cong S_3$, the symmetry group of the ammonia molecule; the tutorial builds that of C_3.

By the end of this lecture you will be able to:

- Compute characters, and use them to find multiplicities and test irreducibility.
- Count irreducible representations and their dimensions.
- Build a small character table.

5.1 Characters and class functions

Can the irreducible content of a representation be read off without finding a single invariant subspace? It can, from traces.

Definition 5.1 (Character)**.** The *character* of a representation (ρ, V) of G is the function

$$\chi_\rho : G \to \mathbb{C}, \qquad \chi_\rho(g) = \operatorname{tr} \rho(g) = \operatorname{tr} D(g).$$

It is well defined, since the trace is the same in every basis. The characters of the irreducible representations $\rho^{(i)}$ are written χ_i and called *irreducible characters*. Note that $\chi_\rho(e) = \operatorname{tr} \mathrm{id}_V = \dim V$, so the character remembers the dimension.

Proposition 5.2 (Basic properties)**.** Let χ be the character of (ρ, V), with G finite. Then:

(i) equivalent representations have equal characters;
(ii) $\chi(hgh^{-1}) = \chi(g)$ for all $g, h \in G$: characters are constant on conjugacy classes;
(iii) $\chi(g^{-1}) = \overline{\chi(g)}$;
(iv) characters add on direct sums: $\chi_{\rho\oplus\sigma} = \chi_\rho + \chi_\sigma$.

Proof. (i) and (ii) are the invariance of the trace under conjugation:

$$\operatorname{tr}(SD(g)S^{-1}) = \operatorname{tr} D(g) \quad \text{and} \quad \operatorname{tr}\big(D(h)D(g)D(h)^{-1}\big) = \operatorname{tr} D(g).$$

(iii) By Lecture 4 we may take $D(g)$ unitary, so

$$D(g^{-1}) = D(g)^{-1} = D(g)^{\dagger},$$

whose trace is the complex conjugate of $\operatorname{tr} D(g)$.

(iv) In a basis adapted to the direct sum the matrices are block-diagonal, and the trace is the sum of the traces of the blocks. ∎

Because characters are constant on conjugacy classes, they lie in a special finite-dimensional space.

Definition 5.3 (Class functions)**.** A *class function* on G is a function $G \to \mathbb{C}$ that is constant on each conjugacy class. Class functions form a vector space $\mathrm{CF}(G)$ whose dimension is the number of conjugacy classes, since each class carries one free value. We give it the inner product

$$\langle \varphi, \psi \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{\varphi(g)}\, \psi(g) = \frac{1}{|G|} \sum_{\text{classes } C} |C|\ \overline{\varphi(g_C)}\, \psi(g_C).$$

In practice one computes with the second form, where g_C is any representative of C.

One more property takes one line to prove. Lecture 7 uses it to count the vibrations of a molecule.

Lemma 5.4 (Fixed-point count)**.** The character of a permutation representation (Lecture 3) counts fixed points:

$$\chi_{\mathrm{perm}}(g) = \#\{\, x \in X : g \cdot x = x \,\}.$$

Proof. The diagonal entries of the permutation matrix are $D_{xx}(g) = \delta_{x,\,g\cdot x}$. So each fixed point contributes a 1. ∎

For S_3 permuting $\{1,2,3\}$, for instance, the identity fixes all three points, a transposition one and a 3-cycle none: $\chi_{\mathrm{perm}} = (3,1,0)$ on the three classes.

5.2 The averaging lemma, and orthogonality

Lecture 4 averaged an inner product over the group. Now we average the operators themselves.

Lemma 5.5 (Averaging lemma)**.** Let (ρ, V) be a representation of a finite group G, and let

$$V^G = \{ v \in V : \rho(g)v = v \text{ for all } g \}$$

be the subspace of invariant vectors. Then $P = \frac{1}{|G|}\sum_{g\in G} \rho(g)$ is a projection onto V^G, and

$$\dim V^G = \frac{1}{|G|} \sum_{g \in G} \chi_V(g).$$

Proof. Fix h. Then $\rho(h)P = \frac{1}{|G|}\sum_g \rho(hg) = P$, since $g \mapsto hg$ only reorders the sum. So every vector Pv is invariant. If v is invariant, then every term of the average is v, and $Pv = v$.

Hence P maps V onto V^G and is the identity there, so $P^2 = P$. The trace of a projection is the dimension of its image (its eigenvalues are 0 and 1, and it is diagonalisable), and $\operatorname{tr} P = \frac{1}{|G|}\sum_g \chi_V(g)$. ∎

On $\mathbb{C}^3$ with S_3 permuting the coordinates, for instance, the only invariant vectors are the multiples of $(1,1,1)$, and indeed $\frac{1}{6}(3 + 3\cdot 1 + 2\cdot 0) = 1$.

The lemma counts invariant vectors. To count *intertwiners*, apply it to the right space. Let V and W be representations, and let $\operatorname{Hom}(V,W)$ be the space of all linear maps $T\colon V \to W$, that is, of all $\dim W \times \dim V$ matrices. The group acts on it by

$$g \cdot T = \rho_W(g)\, T \rho_V(g)^{-1}.$$

This is a representation, since

$$(gh)\cdot T = \rho_W(g)\rho_W(h)\, T \rho_V(h)^{-1}\rho_V(g)^{-1} = g\cdot(h\cdot T).$$

A map T is invariant exactly when $\rho_W(g)\,T = T\rho_V(g)$ for all g, that is, when it is an intertwiner:

$$\operatorname{Hom}(V,W)^G = \operatorname{Hom}_G(V,W).$$

Lemma 5.6. The character of $\operatorname{Hom}(V,W)$ is $g \mapsto \overline{\chi_V(g)}\,\chi_W(g)$.

Proof. Fix g. It acts on matrices T by $T \mapsto ATB$, where $A = D_W(g)$ and $B = D_V(g^{-1})$. The matrix A is diagonalisable: Lecture 4 showed that in a suitable basis it is unitary, and a unitary matrix is diagonalisable. The same holds for B. So we may choose bases of W and V in which $A = \operatorname{diag}(a_1,\dots,a_m)$ and $B = \operatorname{diag}(b_1,\dots,b_n)$. Let E_{kl} be the matrix with a single 1 in row k and column l. Then $AE_{kl}B = a_k b_l\, E_{kl}$, that is, the matrix units are eigenvectors of $T \mapsto ATB$. Its trace is therefore

$$\sum_{k,l} a_k b_l = \Big(\sum_k a_k\Big)\Big(\sum_l b_l\Big) = \operatorname{tr} A\ \operatorname{tr} B = \chi_W(g)\,\chi_V(g^{-1}) = \chi_W(g)\,\overline{\chi_V(g)}.$$

The last step is Proposition 5.2(iii). The trace does not depend on the basis, so this is the character of $\operatorname{Hom}(V,W)$ at g. ∎

Theorem 5.7 (Characters count intertwiners)**.** For representations V and W of a finite group,

$$\langle \chi_V, \chi_W\rangle_G = \dim \operatorname{Hom}_G(V,W).$$

Hence the irreducible characters are orthonormal:

$$\langle \chi_i, \chi_j\rangle_G = \delta_{ij}.$$

Proof. The space $\operatorname{Hom}(V,W)$ is a representation, and its invariant vectors are the intertwiners: $\operatorname{Hom}(V,W)^G = \operatorname{Hom}_G(V,W)$. Applying the averaging lemma to the representation $\operatorname{Hom}(V,W)$, and then Lemma 5.6, gives

$$\begin{aligned}\dim \operatorname{Hom}_G(V,W) = \dim \operatorname{Hom}(V,W)^G &= \frac{1}{|G|}\sum_{g\in G} \chi_{\operatorname{Hom}(V,W)}(g)\\ &= \frac{1}{|G|}\sum_{g\in G} \overline{\chi_V(g)}\,\chi_W(g) = \langle \chi_V, \chi_W\rangle_G .\end{aligned}$$

For irreducibles, Schur's lemma gives $\dim \operatorname{Hom}_G\big(V^{(i)}, V^{(j)}\big) = \delta_{ij}$ (Lecture 4). ∎

The irreducible characters are orthonormal vectors in $\mathrm{CF}(G)$, whose dimension is the number of classes. So *a finite group has at most as many irreducible representations as conjugacy classes*. In other words, every finite group has finitely many irreducible representations.

5.3 Characters determine the representation

Decompose a representation by Maschke's theorem, $V \cong \bigoplus_j m_j V^{(j)}$. Characters add over direct sums, so $\chi_V = \sum_j m_j \chi_j$, and orthonormality picks out each coefficient.

Theorem 5.8 (Multiplicity formula)**.** Let (ρ, V) be any representation, with character χ_V and decomposition $V \cong \bigoplus_j m_j V^{(j)}$. Then

$$m_i = \langle \chi_i, \chi_V \rangle_G .$$

So the multiplicities are inner products, computable from the character alone, with no basis-hunting. Moreover, they do not depend on how V was decomposed.

Proof. $\langle \chi_i, \chi_V \rangle_G = \sum_j m_j \langle \chi_i, \chi_j \rangle_G = m_i$. ∎

Corollary 5.9 (The character is a complete invariant)**.** Two representations of a finite group are equivalent if and only if they have the same character.

Proof. Equivalent representations have equal characters (Proposition 5.2(i)).

Conversely, let (ρ_V, V) and (ρ_W, W) have the same character, $\chi_V = \chi_W$. By Maschke's theorem both decompose into irreducibles:

$$V \cong \bigoplus_i m_i V^{(i)}, \qquad W \cong \bigoplus_i m_i' V^{(i)}.$$

By Theorem 5.8 the multiplicities are fixed by the character alone, so

$$m_i = \langle \chi_i, \chi_V \rangle_G = \langle \chi_i, \chi_W \rangle_G = m_i'$$

for every i. So both representations are equivalent to the same direct sum $U = \bigoplus_i m_i V^{(i)}$: there are invertible intertwiners $S : V \to U$ and $T : W \to U$. Then the map $T^{-1}S : V \to W$ is invertible, and it is an intertwiner too:

$$T^{-1} S \, \rho_V(g) = T^{-1} \rho_U(g) \, S = \rho_W(g) \, T^{-1} S.$$

The middle step is the relation $T\rho_W(g) = \rho_U(g)\,T$ multiplied by T^{-1} on both sides. Hence $V \cong W$. ∎

Corollary 5.10 (Irreducibility criterion)**.** $\langle \chi_V, \chi_V \rangle_G = \sum_i m_i^2$. In particular V is irreducible if and only if $\langle \chi_V, \chi_V \rangle_G = 1$.

Proof. Expand $\chi_V = \sum m_j \chi_j$ in the orthonormal basis and compute the norm. A sum of squares of nonnegative integers equals 1 precisely when one m_j is 1 and the rest vanish. ∎

5.4 The regular representation

Which dimensions can the irreducible representations have? One representation contains them all. The *regular representation* is the permutation representation (Lecture 3) of G acting on itself by left multiplication, $\rho^{\mathrm{reg}}(g)e_h = e_{gh}$, of dimension $|G|$. The equation $gh = h$ has a solution only for $g = e$, so by the fixed-point count

$$\chi_{\mathrm{reg}}(e) = |G| \quad \text{and} \quad \chi_{\mathrm{reg}}(g) = 0 \quad (g \neq e). \tag{5.1}$$

Theorem 5.11 (Sum of squares)**.** Every irreducible representation $V^{(i)}$ occurs in the regular representation, with multiplicity $n_i = \dim V^{(i)}$. Consequently

$$\sum_i n_i^2 = |G| \qquad \text{and} \qquad \sum_i n_i \chi_i(g) = 0 \quad \text{for } g \neq e.$$

Proof. By the multiplicity formula and (5.1),

$$m_i = \langle \chi_i, \chi_{\mathrm{reg}} \rangle_G = \frac{1}{|G|} \overline{\chi_i(e)}\ |G| = n_i.$$

So $\chi_{\mathrm{reg}} = \sum_i n_i \chi_i$. Evaluate at e for the first identity and at $g \neq e$ for the second. ∎

Theorems that once needed structural arguments have become arithmetic. The bound of Section 5.2 is in fact an equality:

Theorem 5.12. The number of inequivalent complex irreducible representations of a finite group equals the number of its conjugacy classes.

We showed above that there are at most as many irreducible representations as classes. It remains to show that there are no fewer, which the self-study section does. What matters most to us is that the irreducible characters form an orthonormal *basis* of the class functions.

5.5 Character tables

All the character theory of a finite group fits in a single square array, the *character table*. It has one row per irreducible character and one column per conjugacy class. The first column is the class of the identity, so it lists the dimensions n_i; the first row is the trivial character, all 1s; the class sizes are recorded above the columns, since the inner product needs them. The rows are orthonormal with weights $|C|\,/|G|$.

To build a table: find the classes (Lecture 2); solve $\sum_i n_i^2 = |G|$ with as many n_i as classes; write down the one-dimensional characters, which are homomorphisms $G \to \mathbb{C} \setminus \{0\}$; fill in the rest from traces of known matrices or from $\sum_i n_i \chi_i(g) = 0$; check orthonormality. For an abelian group such as C_3 every class is a single element and every row is a χ_k (Lecture 4); the tutorial writes out that table.

[tutorial] **Example 5.13** (C_3)**.** For the abelian group C_n every element is its own class, and the n irreducible characters are the $\chi_k(r^m) = \omega^{km}$, $\omega = e^{2\pi \mathrm{i}/n}$ (Lecture 4). For $n = 3$:

C_3	e	r	r^2
χ_0	1	1	1
χ_1	1	ω	ω^2
χ_2	1	ω^2	ω

Orthogonality of rows is the geometric-series identity $1+\omega^k+\omega^{2k}=0$ for $k \not\equiv 0$. So character theory on cyclic groups *is* the arithmetic of roots of unity, the discrete Fourier transform written as a table. The two complex characters χ_1 and $\chi_2 = \overline{\chi_1}$ will return in Lecture 6. ◁

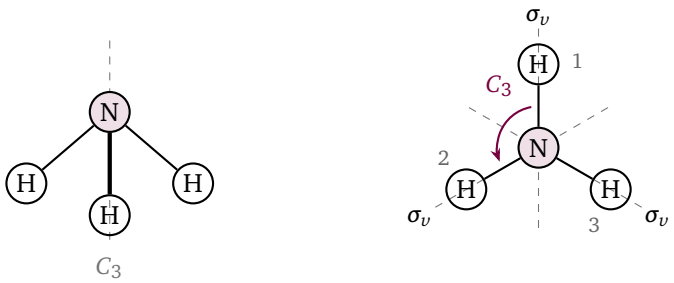


Figure 5.1: The ammonia molecule NH_3. On the left it is seen from the side (the thick bond points towards the reader), on the right from above, down the axis C_3. The dashed lines on the right are the three mirror planes σ_v, and the arrow is the rotation through $2\pi/3$, which takes hydrogen 1 to the place of hydrogen 2.

Example 5.14 ($C_{3v} \cong S_3$: the ammonia table)**.** The ammonia molecule NH_3 is a flat-bottomed pyramid: three hydrogens at the corners of an equilateral triangle, the nitrogen on the axis above its centre (Figure 5.1). Its symmetry operations are the identity, the rotations through $\pm 2\pi/3$ about the axis, and the reflections in the three vertical planes that each contain the axis and one hydrogen. They permute the three hydrogens as S_3 permutes $\{1,2,3\}$, and the group, written C_{3v} (Lecture 7 explains the name), is isomorphic to $D_3 \cong S_3$.

Its classes are the identity E, the two rotations $2C_3$ and the three reflections $3\sigma_v$ (Lecture 2), of sizes $1, 2, 3$. Three classes and $|G| = 6$ force the dimensions $1 + 1 + 4 = 6$: two one-dimensional representations and one two-dimensional.

The one-dimensional ones are the trivial character and the sign of the permutation, which is $+1$ on the rotations and -1 on the reflections. The two-dimensional row follows from $\sum_i n_i \chi_i(g) = 0$: on C_3, $1 + 1 + 2\chi(C_3) = 0$ gives $\chi(C_3) = -1$, and on σ_v, $1 - 1 + 2\chi(\sigma_v) = 0$ gives $\chi(\sigma_v) = 0$. These are the traces of the defining representation of D_3 (Lecture 3): $2\cos\frac{2\pi}{3} = -1$ for a rotation and 0 for a reflection, whose eigenvalues are 1 and -1.

To see how the characters come out of matrices, write down the matrices of all three representations for every group element. Here σ_j is the reflection in the plane containing hydrogen j. The matrices $D_E(g)$ of E are written in the basis of Lecture 3, with hydrogen 1 on the y-axis (Figure 5.1, right):

C_{3v}	E	C_3	C_3^2	σ_1	σ_2	σ_3
A_1	1	1	1	1	1	1
A_2	1	1	1	-1	-1	-1
E	$\begin{pmatrix}1&0\\0&1\end{pmatrix}$	$\frac{1}{2}\begin{pmatrix}-1&-\sqrt{3}\\\sqrt{3}&-1\end{pmatrix}$	$\frac{1}{2}\begin{pmatrix}-1&\sqrt{3}\\-\sqrt{3}&-1\end{pmatrix}$	$\begin{pmatrix}-1&0\\0&1\end{pmatrix}$	$\frac{1}{2}\begin{pmatrix}1&\sqrt{3}\\\sqrt{3}&-1\end{pmatrix}$	$\frac{1}{2}\begin{pmatrix}1&-\sqrt{3}\\-\sqrt{3}&-1\end{pmatrix}$

A one-dimensional matrix is a number, and that number is its own trace. The traces of the two-dimensional matrices are

$$\operatorname{tr} D_E(E) = 2, \qquad \operatorname{tr} D_E(C_3) = \operatorname{tr} D_E(C_3^2) = -\tfrac{1}{2} - \tfrac{1}{2} = -1, \qquad \operatorname{tr} D_E(\sigma_j) = 0.$$

The matrices $D_E(C_3)$ and $D_E(C_3^2)$ differ, yet their traces agree, and the same holds for the three reflections. So it must be: a character is constant on a class (Proposition 5.2(ii)), even

where the matrices are not. One trace per class gives the character table:

C_{3v}	E	$2C_3$	$3\sigma_v$	
A_1	1	1	1	z
A_2	1	1	-1	
E	2	-1	0	(x, y)

Check: $\langle \chi_E, \chi_E \rangle_G = \frac{1}{6}(4 + 2 \cdot 1 + 3 \cdot 0) = 1$, so the defining representation is irreducible, as the tutorial of Lecture 4 showed by hand.

The labels are Mulliken's, which chemists and spectroscopists use. A and B are one-dimensional, even and odd under the principal rotation; E is two-dimensional and T three-dimensional; subscripts and primes tell apart representations of the same kind. The representation E has nothing to do with the column E, the identity operation: both names are entrenched, and the context always tells which is meant. The last column records which coordinate functions carry each representation, with z along the threefold axis: z is unchanged by every operation, and (x, y) carry the defining representation, as Lecture 3 showed. ◁

Now the decomposition that Lecture 4's tutorial did by hand takes one line. The permutation character $\chi_{\text{perm}} = (3, 0, 1)$ on $E, 2C_3, 3\sigma_v$ gives

$$m_{A_1} = \tfrac{1}{6}(3 + 0 + 3) = 1, \qquad m_{A_2} = \tfrac{1}{6}(3 + 0 - 3) = 0, \qquad m_E = \tfrac{1}{6}(6 + 0 + 0) = 1,$$

so $\mathbb{C}^3 \cong A_1 \oplus E$, and the absence of A_2 is seen at once. As a cross-check,

$$\langle \chi_{\text{perm}}, \chi_{\text{perm}} \rangle_G = \tfrac{1}{6}(9 + 0 + 3) = 2 = 1^2 + 1^2,$$

so there are exactly two irreducible constituents.

[tutorial] **Example 5.15** (D_4)**.** The classes (Lecture 2) are

$$\{e\}, \quad \{r^2\}, \quad \{r, r^3\}, \quad \{s, r^2 s\}, \quad \{rs, r^3 s\}.$$

Their sizes are $1, 1, 2, 2, 2$. The dimension pattern $(1, 1, 1, 1, 2)$ is forced: five squares summing to 8, one of them 1. Take a one-dimensional character ν. Since r is conjugate to $r^{-1} = r^3$, $\nu(r) = \nu(r)^{-1}$, i.e. $\nu(r) = \pm 1$. From $s^2 = e$ we get $\nu(s) = \pm 1$. All four sign choices are consistent with the relations, and always $\nu(r^2) = \nu(r)^2 = 1$. The two-dimensional row follows from $\sum_i n_i \chi_i(c) = 0$ for each class $c \neq \{e\}$: $\chi_5(c) = -\frac{1}{2} \sum_{1\text{-dim}} \nu(c)$, giving $\chi_5(r^2) = -2$ and $\chi_5(r) = \chi_5(s) = \chi_5(rs) = 0$:

D_4	$1\,e$	$1\,r^2$	$2\,r$	$2\,s$	$2\,rs$
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	1	-1	1	-1
χ_4	1	1	-1	-1	1
χ_5	2	-2	0	0	0

As a check, the defining representation $D(r) = R(\pi/2)$, $D(s) = \operatorname{diag}(-1, 1)$ has traces $2, -2, 0, 0, 0$. So its character *is* χ_5, and $\langle \chi_5, \chi_5 \rangle_G = \frac{1}{8}(4 + 4 + 0 + 0 + 0) = 1$ confirms irreducibility. The quaternion group Q_8 is not isomorphic to D_4, yet it has an *identical* character table (Exercise Sheet 5). So the character table is a 'strong' invariant, but it does not determine the group. ◁

5.6 More orthogonality *[self-study]*

As many irreducibles as classes. It remains to show that the irreducible characters span the class functions. Let f be a class function orthogonal to every irreducible character. For each irreducible $(\rho_i, V^{(i)})$, the operator $A_i = \sum_g \overline{f(g)}\,\rho_i(g)$ commutes with every $\rho_i(h)$, because f is constant on classes, so renaming $g \mapsto hgh^{-1}$ leaves the sum unchanged. By Schur's lemma $A_i = \lambda\,\mathrm{id}$. Take the trace of both sides: $\operatorname{tr}(\lambda\,\mathrm{id}) = \lambda\, n_i$, and $\operatorname{tr}\rho_i(g) = \chi_i(g)$. Using the definition of the inner product (Definition 5.3),

$$\langle f, \chi_i\rangle_G = \frac{1}{|G|}\sum_g \overline{f(g)}\,\chi_i(g),$$

we get

$$\lambda\, n_i = \sum_g \overline{f(g)}\,\chi_i(g) = |G|\ \langle f, \chi_i\rangle_G = 0.$$

The last equality is the assumption that f is orthogonal to χ_i. Since $n_i \neq 0$, $\lambda = 0$. So $A_i = 0$ for every irreducible, hence $\sum_g \overline{f(g)}\,\rho(g) = 0$ on every representation, by Maschke. In the regular representation, apply this operator to the basis vector e_e: it gives $\sum_g \overline{f(g)}\,e_g = 0$, so $f = 0$. The irreducible characters are therefore a basis of the class functions, and their number is the number of classes. This proves Theorem 5.12.

Theorem 5.16 (Second orthogonality relation)**.** For any $g, h \in G$,

$$\sum_i \overline{\chi_i(g)}\,\chi_i(h) = \begin{cases} \dfrac{|G|}{|[g]|} = |C_G(g)|\,, & h \text{ conjugate to } g,\\[2mm] 0, & \text{otherwise,}\end{cases}$$

the sum running over all irreducible characters.

Proof. Let $C_1, \dots, C_k$ be the classes and form the matrix $X_{iC} = \sqrt{|C|\,/|G|}\;\chi_i(g_C)$, one row for each irreducible character and one column for each class. It is square by Theorem 5.12. The first orthogonality relation says precisely that the rows of X are orthonormal, $XX^\dagger = \mathbb{1}$. A square matrix with orthonormal rows is unitary, so its columns are orthonormal too, $X^\dagger X = \mathbb{1}$, which is the assertion. The centraliser form follows from the identity $|[g]| = [G : C_G(g)]$ of Lecture 2. ∎

For $g = h = e$ the relation is $\sum_i n_i^2 = |G|$ again, and for $g = e \neq h$ it is the identity $\sum_i n_i\,\chi_i(h) = 0$ used to build the tables.

The great orthogonality theorem. Not only the traces but the matrix elements are orthogonal. If $D^{(i)}$ are unitary matrices of the inequivalent irreducibles, then

$$\frac{1}{|G|}\sum_g \overline{D^{(i)}_{\mu\nu}(g)}\,D^{(j)}_{\alpha\beta}(g) = \frac{1}{n_i}\,\delta_{ij}\,\delta_{\mu\alpha}\,\delta_{\nu\beta}.$$

The proof averages $\rho^{(j)}(g)\,E\,\rho^{(i)}(g)^{-1}$ over the group for a matrix unit E, which gives an intertwiner, and applies Schur's lemma: zero for $i \neq j$, a scalar fixed by its trace for $i = j$. Summing over $\mu = \nu$ and $\alpha = \beta$ recovers Theorem 5.7.

Fourier analysis on a finite group. Counting, $\sum_i n_i^2 = |G|$ matrix elements $\sqrt{n_i}\,D^{(i)}_{\mu\nu}$ are orthonormal functions on G, which has $|G|$ elements. So they are a basis of all functions on G: every function has a unique expansion in matrix elements of the irreducibles. For

C_n the matrix elements are the $\chi_k(r^m) = e^{2\pi i k m/n}$, and the expansion is the discrete Fourier transform. Part II meets the same expansion for U(1), where it is the Fourier series, and for the rotation group, where it produces the spherical harmonics (the Peter–Weyl theorem).

The dual representation of Lecture 3 has the conjugate character: $D^*(g) = D(g^{-1})^{\mathsf{T}}$, and transposition does not change the trace, so $\chi_{\rho^*}(g) = \chi_\rho(g^{-1}) = \overline{\chi_\rho(g)}$.

5.7 Appendix: the regular representation of the ammonia group

[self-study]

By Theorem 5.11 the regular representation of C_{3v} decomposes as

$$\mathbb{C}^6 \cong A_1 \oplus A_2 \oplus 2E, \qquad 6 = 1 + 1 + 2 \cdot 2.$$

Let us see this decomposition in matrices. We find a single change of basis P that turns all six matrices $\rho^{\mathrm{reg}}(g)$ at once into block-diagonal ones, with unitary blocks.

Order the basis vectors e_h as $h = E, C_3, C_3^2, \sigma_1, \sigma_2, \sigma_3$, as in the table of matrices in Example 5.14. The matrix $\rho^{\mathrm{reg}}(g)$ takes e_h to e_{gh}, so its column h has a one in row gh. The products gh come from multiplying the matrices D_E. For instance, $C_3\sigma_1 = \sigma_3$, since $D_E(C_3)D_E(\sigma_1) = D_E(\sigma_3)$. The group is generated by C_3 and σ_1, whose matrices are

$$\rho^{\mathrm{reg}}(C_3) = \begin{pmatrix} 0&0&1&0&0&0\\ 1&0&0&0&0&0\\ 0&1&0&0&0&0\\ 0&0&0&0&1&0\\ 0&0&0&0&0&1\\ 0&0&0&1&0&0 \end{pmatrix}, \qquad \rho^{\mathrm{reg}}(\sigma_1) = \begin{pmatrix} 0&0&0&1&0&0\\ 0&0&0&0&1&0\\ 0&0&0&0&0&1\\ 1&0&0&0&0&0\\ 0&1&0&0&0&0\\ 0&0&1&0&0&0 \end{pmatrix}.$$

The columns of P are built from the matrix elements of the irreducibles. For each irreducible $D^{(i)}$ and each pair of indices μ, ν, take the vector

$$w^{(i)}_{\mu\nu} = \sqrt{\frac{n_i}{|G|}} \sum_{h \in G} \overline{D^{(i)}_{\mu\nu}(h)}\, e_h .$$

By the great orthogonality theorem (Section 5.6) these $\sum_i n_i^2 = |G|$ vectors are orthonormal. For fixed ν, the vectors $w^{(i)}_{1\nu}, \dots, w^{(i)}_{n_i\nu}$ transform as a basis of $D^{(i)}$:

$$\rho^{\mathrm{reg}}(g)\, w^{(i)}_{\mu\nu} = \sum_{\kappa} D^{(i)}_{\kappa\mu}(g)\, w^{(i)}_{\kappa\nu} .$$

Indeed, $\rho^{\mathrm{reg}}(g)$ moves the coefficient of e_h to e_{gh}. So the coefficient of e_h in $\rho^{\mathrm{reg}}(g)\, w^{(i)}_{\mu\nu}$ is $\overline{D^{(i)}_{\mu\nu}(g^{-1}h)}$ times $\sqrt{n_i/|G|}$, and unitarity, $D^{(i)}(g^{-1}) = D^{(i)}(g)^\dagger$, gives

$$\overline{D^{(i)}_{\mu\nu}(g^{-1}h)} = \sum_{\kappa} \overline{D^{(i)}_{\mu\kappa}(g^{-1})}\; \overline{D^{(i)}_{\kappa\nu}(h)} = \sum_{\kappa} D^{(i)}_{\kappa\mu}(g)\, \overline{D^{(i)}_{\kappa\nu}(h)} .$$

So each $\nu = 1, \dots, n_i$ gives one copy of $D^{(i)}$, and $D^{(i)}$ occurs n_i times, as Theorem 5.11 says.

All the matrices of C_{3v} are real, so complex conjugation changes nothing. The columns of P are, in order, $w_{11}^{(A_1)}, w_{11}^{(A_2)}, w_{11}^{(E)}, w_{21}^{(E)}, w_{12}^{(E)}, w_{22}^{(E)}$:

$$P = \frac{1}{2\sqrt{3}} \begin{pmatrix} \sqrt{2} & \sqrt{2} & 2 & 0 & 0 & 2 \\ \sqrt{2} & \sqrt{2} & -1 & \sqrt{3} & -\sqrt{3} & -1 \\ \sqrt{2} & \sqrt{2} & -1 & -\sqrt{3} & \sqrt{3} & -1 \\ \sqrt{2} & -\sqrt{2} & -2 & 0 & 0 & 2 \\ \sqrt{2} & -\sqrt{2} & 1 & \sqrt{3} & \sqrt{3} & -1 \\ \sqrt{2} & -\sqrt{2} & 1 & -\sqrt{3} & -\sqrt{3} & -1 \end{pmatrix}.$$

For instance, the third column $w_{11}^{(E)}$ is the upper-left entries $1, -\frac{1}{2}, -\frac{1}{2}, -1, \frac{1}{2}, \frac{1}{2}$ of the matrices $D_E(h)$, multiplied by $\sqrt{2/6} = 1/\sqrt{3}$. The columns are orthonormal, so $P^{-1} = P^{\mathsf{T}}$, and for every g

$$P^{-1}\rho^{\mathrm{reg}}(g)\,P = \begin{pmatrix} D_{A_1}(g) & & & \\ & D_{A_2}(g) & & \\ & & D_E(g) & \\ & & & D_E(g) \end{pmatrix},$$

where $D_{A_1}(g) = 1$ and $D_{A_2}(g) = \pm 1$ is the sign representation. For the generators,

$$P^{-1}\rho^{\mathrm{reg}}(C_3)\,P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}, \qquad P^{-1}\rho^{\mathrm{reg}}(\sigma_1)\,P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

The block $D_E(g)$ appears twice, because E occurs twice in the regular representation. Both matrices have trace zero,

$$1 + 1 - \tfrac{1}{2} - \tfrac{1}{2} - \tfrac{1}{2} - \tfrac{1}{2} = 0, \qquad 1 - 1 - 1 + 1 - 1 + 1 = 0,$$

as (5.1) requires. One change of basis P serves all six elements at once: this is Maschke's theorem, written out in numbers.

Summary.

- *Characters* $\chi_V(g) = \operatorname{tr}\rho(g)$ are class functions, with $\chi_V(e) = \dim V$; a permutation character counts fixed points (Proposition 5.2, Lemma 5.4).
- *Averaging lemma.* $\frac{1}{|G|}\sum_g \rho(g)$ projects onto the invariant vectors, whose number is the average of χ_V (Lemma 5.5). Applied to $\operatorname{Hom}(V,W)$ it gives $\langle \chi_V, \chi_W\rangle_G = \dim \operatorname{Hom}_G(V,W)$, so irreducible characters are orthonormal (Theorem 5.7).
- *Multiplicities.* $m_i = \langle \chi_i, \chi_V\rangle_G$; the character determines the representation; V is irreducible exactly when $\langle \chi_V, \chi_V\rangle_G = 1$ (Theorem 5.8 and its corollaries).
- *Counting.* $\sum_i n_i^2 = |G|$, from the regular representation (Theorem 5.11), and there are as many irreducibles as classes (Theorem 5.12).
- *Tables.* $C_{3v} \cong S_3$ has A_1, A_2, E, and $\mathbb{C}^3 \cong A_1 \oplus E$ in one line.

Tutorial. Exercise Sheet 5.

References

Serre, *Linear Representations of Finite Groups*, Ch. 2 (character theory); Jones, *Groups, Representations and Physics*, §§4.2–4.4; Tung, *Group Theory in Physics*, §§3.5–3.7.